Roger Torres

484-274-9262 | rogertorressa@gmail.com | linkedin.com/in/rogertorresro | github.com/rTorresro | rtorresro.github.io

Education

University of Pittsburgh — Pittsburgh, PA
Bachelor of Science in Computer Science — *Expected December 2028*
- **Relevant Coursework:** Data Structures & Algorithms, Computer Organization & Assembly Language, Mathematical Foundations of Machine Learning, Discrete Mathematics, Linear Algebra

Experience

Software Development Intern — Canonsburg, PA (Hybrid)
Synopsys — *August 2026 – December 2026*
- Incoming intern on the software development team at Synopsys, a leading provider of electronic design automation (EDA) software.

Undergraduate Teaching Assistant, CS 0445 — Pittsburgh, PA
University of Pittsburgh — *August 2026 – December 2026*
- Selected as UTA for CS 0445 (Data Structures & Algorithms), supporting 100+ students through office hours, grading, and course material.

CS Peer Tutor — Pittsburgh, PA
University of Pittsburgh — *August 2026 – December 2026*
- Provide one-on-one and small-group tutoring support to Computer Science students.

Projects

Pulseboard | *Spring Boot, React, TypeScript, PostgreSQL*
- Built a productivity analytics dashboard with 10+ REST endpoints via Spring Boot, tracking tasks and visualizing streaks, weekly totals, and completion rates.
- Modeled relational data with PostgreSQL and JPA; implemented JWT-based auth with user-scoped data access and sub-200ms aggregation queries.
- Developed deadline comparison logic that flagged 85% of at-risk tasks before due dates via automated alerts.

Queued | *Node.js, Express, React, TypeScript, TMDB API*
- Built a full-stack watchlist platform integrating the TMDB API to search and catalog 800,000+ movies and TV shows with real-time metadata and personalized viewing history.
- Engineered a decision-helper filtering by runtime, genre, and mood — cutting average selection time to under 2 minutes across 15+ genre categories.
- Implemented user authentication with session-scoped watchlist management, analytics dashboards, and viewing trend visualizations.

KeyGuide | *React, TypeScript, Tone.js, Vite, Tailwind CSS, Python*
- Built a browser-based piano learning app with a 3-octave virtual keyboard, 3 practice modes (auto-play, step-through, MIDI wait-for-correct), and a searchable song library with difficulty ratings.
- Integrated Web MIDI API for real keyboard input with live visual feedback and Tone.js PolySynth for low-latency audio synthesis across 36 keys.
- Built a Python CLI pipeline using mido to convert MIDI files into structured song JSON with auto-detected hand splits and heuristic fingering assignments.

RunIt Pittsburgh | *React, Vite, Firebase, Leaflet, Tailwind CSS*
- Built a web app connecting basketball players to pickup games in real-time, mapping 25+ courts across Pittsburgh with Firebase-backed live check-ins.
- Developed an interactive Leaflet map with real-time court markers and player counts; achieved sub-1.5s page loads with Vite and responsive design via Tailwind CSS.

Technical Skills

Languages: Java, Python, C++, JavaScript, TypeScript, HTML, CSS
Libraries & Frameworks: React, Node.js, Express, Spring Boot, Tailwind CSS, JUnit, Tone.js
Databases & Cloud: PostgreSQL, Firebase
Developer Tools: Git, GitHub, Maven, VS Code, IntelliJ, Linux

Leadership & Extracurriculars

Pitt Computer Science Club — University of Pittsburgh
Member — *August 2025 – Present*
- Won the Pitt CSC × CGI Coding Competition, awarded $250 in prizes, competing against fellow CS students in a university-sponsored challenge.
- Participate in weekly meetings, workshops, and speaker events.